0.1 Preliminaries

Note: $(\mathbb{Z}_n, +, \cdot)$ is a ring. If $n = p$ is a prime, then for any non-zero $a \in \mathbb{Z}_p$, by Fermat's Little Theorem:

$$a^{p-1} \equiv 1 \pmod{p}$$

That is, the multiplicative inverse of a is given by $a^{-1} = a^{p-2} \pmod{p} \in \mathbb{Z}_p$.

Proposition 0.1.1. Let F be a field.

- If $\mathrm{ch}(F) = 0$, then F must contain a subfield isomorphic to $\mathbb{Q}$.
- If $\mathrm{ch}(F) = p \neq 0$, then F must contain a subfield isomorphic to $\mathbb{F}_p$, where $p = \mathrm{ch}(F)$.

Proposition 0.1.2. If $\deg_{\mathbb{F}_p} \alpha = n$, then the simple extension $F(\alpha)$ has p^n elements, because $F(\alpha) \cong \mathbb{F}_{p^n}$. Furthermore, if E is a finite field, then E has p^n elements for some $n \in \mathbb{N}$, where $p = \mathrm{ch}(E)$. This follows from the fact that since E is finite, $\dim_{\mathbb{F}_p} E$ must be finite.

Theorem 0.1.3. Let $\mathbb{Z}_p \leq E \leq \overline{\mathbb{Z}_p}$ be a field with p^n elements. Then, for the polynomial $P(x) = x^{p^n} - x \in \mathbb{Z}_p[x]$,

$$E = \{\alpha \in \overline{\mathbb{Z}_p} \mid P(\alpha) = 0\}.$$

Proof. Since $(E \setminus \{0\}, \times)$ is a multiplicative group of order $p^n - 1$, for any $\alpha \in E \setminus \{0\}$, the order of α divides $p^n - 1$. Thus, $\alpha^{p^n - 1} = 1$, which implies $\alpha^{p^n} - \alpha = 0$, so $P(\alpha) = 0$. Clearly $P(0) = 0$ as well. Since the polynomial $P(x)$ can have at most p^n zeros in $\overline{\mathbb{Z}_p}$, and we have found p^n distinct zeros in E, we obtain the result. $\square$

0.2 The Frobenius Automorphism

Theorem 0.2.1. Let F be a finite field of characteristic p. Then, the map

$$\varphi_p : F \to F; \quad x \mapsto x^p$$

is an automorphism (the Frobenius Automorphism). Moreover, the fixed field $F_{\{\varphi_p\}}$ is $\mathbb{Z}_p$.

Proof. **Well-definedness** is clear.
Homomorphism: For any $a, b \in F$,

$$\varphi_p(a+b) = (a+b)^p = \sum_{k=0}^{p} \binom{p}{k} a^{p-k} b^k = a^p + b^p = \varphi_p(a) + \varphi_p(b)$$

since $\mathrm{ch}(F) = p$ (all intermediate binomial coefficients are multiples of p). Multiplication is clearly preserved: $\varphi_p(ab) = (ab)^p = a^p b^p = \varphi_p(a)\varphi_p(b)$.
Injectivity: $\varphi_p(a) = a^p = 0 \iff a = 0$, thus $\ker \varphi_p = \{0\}$.
Surjectivity: A finite injective endomorphism is necessarily surjective.
Meanwhile, by Fermat's Little Theorem, for any $a \in \mathbb{Z}_p$, $\varphi_p(a) = a^p = a$.
Thus $\mathbb{Z}_p \subseteq F_{\{\varphi_p\}}$. Since $x^p - x = 0$ has at most p zeros in F, we conclude $F_{\{\varphi_p\}} = \mathbb{Z}_p$. □

Corollary 0.2.2. If $\mathrm{ch}(F) = p$, then $(a+b)^{p^n} = a^{p^n} + b^{p^n}$ for any $a, b \in F$ and $n \geq 1$.

0.3 Existence of Finite Fields

> **Theorem 0.3.1.** If F is a field with $\mathrm{ch}(F) = p$, then $x^{p^n} - x \in F[x]$ has p^n distinct zeros in $\overline{F}$.

Proof. Observe that $x^{p^n} - x = x(x^{p^n-1} - 1)$. Let $f(x) = x^{p^n-1} - 1$. Suppose $\alpha \in \overline{F}$ is a root of $f(x)$. Then

$$f(x) = (x - \alpha)(x^{p^n-2} + \alpha x^{p^n-3} + \cdots + \alpha^{p^n-3}x + \alpha^{p^n-2})$$

Let $g(x) = f(x)/(x - \alpha)$. Then,

$$g(\alpha) = \underbrace{1 + 1 + \cdots + 1}_{p^n-1 \text{ terms}} \cdot \alpha^{p^n-2} = (p^n - 1) \cdot \alpha^{-1} = -\alpha^{-1}$$

since $p \cdot 1 = 0$ in characteristic p. Since $\alpha \neq 0$, we have $-\alpha^{-1} \neq 0$. This means α is a zero of f of multiplicity 1. Therefore, all zeros are distinct. □

> **Theorem 0.3.2.** A finite field of order p^n exists for every prime p and $n \in \mathbb{N}$.

Proof. Let $K = \{x \in \overline{\mathbb{Z}_p} \mid x^{p^n} - x = 0\}$. By the previous theorem, K contains p^n distinct elements. It remains to show K is a field. If $x, y \in K$, then $(x \pm y)^{p^n} = x^{p^n} \pm y^{p^n} = x \pm y$, so $x \pm y \in K$. Similarly, $(xy)^{p^n} = x^{p^n}y^{p^n} = xy$ and $(x/y)^{p^n} = x/y$, so K is closed under field operations. Thus K is a subfield of $\overline{\mathbb{Z}_p}$ with p^n elements. □